

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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**Project Name: HireFlow — AI-Powered HR Hiring
Pipeline**

Group Name: We googled everything

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Project Name: HireFlow

1. Project Overview

- **Problem Statement:** Traditional hiring processes are often slow, repetitive, and inefficient. HR teams spend significant time manually screening CVs, coordinating interview invitations, evaluating candidates, sending follow-up emails, and tracking applicants across multiple stages. Small and medium-sized companies especially struggle with limited HR manpower, inconsistent evaluation standards, delayed communication, and poor candidate experience. Existing hiring tools may lack intelligent automation, seamless workflow integration, or affordable AI-driven decision support.
- **Target Domain:** AI-powered recruitment management and agentic workflow automation for modern hiring pipelines, enabling organizations to streamline talent acquisition from job posting to onboarding.
- **Proposed Solution Summary:** HireFlow reduces hiring friction by automating key recruitment tasks through an intelligent end-to-end platform. It assists HR teams with AI CV screening, automated interview invitations, AI interview scoring, candidate ranking, email notifications, and offer letter generation—all within a centralized dashboard. By combining workflow automation with AI reasoning, HireFlow transforms HR teams from manual process operators into strategic talent decision-makers, while improving efficiency, consistency, and candidate experience.

2. Background & Business Objective

- **Background of the Problem:** Traditional hiring workflows still rely heavily on manual and fragmented processes. HR teams often need to review large volumes of CVs one by one, coordinate interview schedules through back-and-forth communication, follow up with applicants, compare candidates manually, and prepare offer-related communication. These repetitive tasks consume valuable time and can slow down recruitment cycles, especially when multiple vacancies or high application volumes are involved. Inconsistent evaluation methods and limited visibility across hiring stages can also make decision-making less efficient and less standardized.
- **Importance of Solving This Issue:** Improving the hiring process can significantly reduce administrative burden, shorten recruitment timelines, and create a smoother candidate experience. Faster screening and ranking allows organizations to secure top talent before competitors. Standardized AI-assisted evaluations can improve fairness and consistency, while automated communication ensures candidates receive timely updates. By reducing operational bottlenecks, HR teams can focus more on strategic talent decisions, employer branding, and workforce growth.
- **Strategic Fit / Impact:** HireFlow aligns strongly with the growing adoption of AI-driven business automation and intelligent workflow systems. It demonstrates how AI can be embedded into real operational processes rather than used only as a standalone chatbot or analytics tool. For organizations, HireFlow offers a scalable recruitment pipeline that can support multiple job openings and candidate pools simultaneously. For the broader market, it reflects the future of HR technology where human decision-makers remain in control while AI handles repetitive execution tasks with speed and consistency.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-assisted hiring management platform that streamlines recruitment workflows, automates repetitive HR tasks, and supports smarter candidate evaluation while keeping final hiring decisions under human control.

3.2. Intended Users (Target Audience):

- Human Resource (HR) Teams
- Talent Acquisition Specialists
- Recruiters
- Hiring Managers
- Small and Medium Enterprises (SMEs) with limited HR manpower
- Startups with growing recruitment needs
- Companies managing multiple job openings simultaneously
- Job Candidates applying through public hiring portals

4. System Functionalities

4.1. Description: HireFlow operates as an AI-assisted recruitment pipeline platform that combines HR workflow management, candidate tracking, and intelligent automation in one centralized system. The platform streamlines the full hiring lifecycle—from job posting and candidate application to AI screening, interview evaluation, ranking, offer generation, and onboarding. By integrating automation into repetitive hiring steps, HireFlow enables HR teams to focus on final decision-making while maintaining transparency, auditability, and control.

4.2. Key Functionalities:

- **HR Authentication:** Secure HR registration and login using email/password, Google OAuth support, JWT-protected dashboard access, and authenticated session management.
- **Job Management:** Create, edit, close, and manage job postings with customizable requirements, department, location, and AI screening settings such as thresholds and shortlist size.

- **AI CV Screening:** Automatically evaluates uploaded CVs against job requirements, producing scores, strengths, weaknesses, and recommendations such as PASS, REVIEW, or REJECT. HR can override AI decisions when needed.
- **AI Interview Engine:** Invites shortlisted candidates through secure token links to complete AI-generated interviews consisting of DSA, MCQ, and behavioral questions. Responses are evaluated automatically.
- **Proctoring & Integrity Monitoring:** Tracks suspicious behavior during interviews such as tab switching, camera loss, and copy-paste attempts. Integrity penalties may reduce final interview scores.
- **Candidate Ranking & Shortlisting:** Ranks applicants using weighted scoring formulas combining CV score, technical tests, behavioral responses, and proctoring results. HR can shortlist or reject candidates manually.
- **Human Interview Workflow:** Schedules physical or online interviews with email confirmations, reschedule requests, and final hiring decisions after interview completion.
- **Offer Letter & Onboarding:** Generates AI-assisted offer letter content for selected candidates, sends acceptance emails, and transitions candidates into onboarding or hired stages.
- **Dashboard & Audit Trail:** Provides funnel analytics, candidate pipeline visibility, workflow logs, and status history for accountability and decision tracking.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: HireFlow uses Z AI GLM as the core large language model for reasoning-based recruitment tasks. GLM was selected because it provides strong natural language understanding, structured output generation, evaluation capabilities, and

cost-effective inference suitable for hackathon-scale deployment. Its strengths in summarization, comparison, reasoning, and text generation make it effective for CV screening, candidate scoring, interview analysis, and offer letter drafting.

4.3.2. Prompting Strategy: The system adopts a multi-step agentic prompting strategy combined with structured zero-shot prompts. Instead of relying on a single prompt, HireFlow separates tasks into smaller workflows such as CV extraction, requirement matching, score generation, recommendation logic, and summary generation. This improves consistency, reduces ambiguity, and ensures outputs are aligned to hiring needs. Multi-step prompting is ideal because recruitment decisions require multiple reasoning layers rather than one simple response.

4.3.3. Context & Input Handling: HireFlow accepts unstructured inputs such as CV PDFs, pasted resumes, interview responses, and job descriptions. Inputs are first cleaned and converted into structured text before being sent to the model. For oversized documents, the system uses truncation or chunking strategies—splitting content into relevant sections such as education, experience, and skills. Priority is given to the most relevant job-matching sections. Extremely large or corrupted files are rejected gracefully with user feedback requesting re-upload.

4.3.4. Fallback & Failure Behavior: When the model returns off-topic, incomplete, or hallucinated responses, HireFlow applies validation rules to detect malformed outputs or missing fields. The system can retry prompts automatically with stricter formatting instructions. If AI confidence is low, candidates are marked for manual HR review instead of automatic rejection. In critical decisions, HR always has override authority. If model services fail entirely, the platform falls back to manual workflow mode so hiring operations can continue without disruption.

5. User Stories & Use Cases

- **User Stories:**

- An HR user wants to register and log in securely so that they can safely access the hiring dashboard and manage recruitment activities. An HR user wants to create job postings so that candidates are able to apply for available roles through the platform. An HR user wants to configure AI screening thresholds so that the strictness of candidate evaluation can be adjusted according to different job requirements.
- An HR user wants to review AI-generated CV analysis so that screening decisions can be made faster and with better insights. An HR user wants to accept, reject, or override AI recommendations so that final hiring authority always remains under human control. An HR user wants shortlisted candidates to automatically receive AI interview invitations so that early-stage interview processes can be streamlined and managed efficiently.
- An HR user wants to view candidate rankings and interview scores so that applicants can be compared consistently using structured evaluation data. An HR user wants to schedule human interviews so that qualified candidates can move forward into the final stage of assessment. An HR user wants to send offer letters automatically so that successful candidates receive professional and timely communication.
- A candidate wants to apply through a public job link so that they can submit their CV easily without the need to create an account. A candidate wants to receive interview instructions by email so that they clearly understand the next steps in the hiring process. A candidate wants to complete the AI interview online so that they can be evaluated remotely in a convenient manner. A candidate wants to confirm or request rescheduling for a human interview so that they can coordinate their availability with the hiring team.

- **Use Cases (Main Interactions):**

Job Posting Management: HR logs into the dashboard, creates a new job posting, sets requirements, and configures AI screening settings such as pass threshold and shortlist size.

Candidate Application Flow: Candidate opens the public apply link, fills in personal information, uploads a CV, and submits the application into the hiring pipeline.

AI CV Screening Flow: The system extracts CV content, compares it against job requirements, generates AI scores, strengths, weaknesses, and recommends PASS, REVIEW, or REJECT. HR may override results.

AI Interview Flow: Selected candidates receive a unique tokenized interview link. Candidates complete technical, MCQ, and behavioral questions. The system evaluates responses automatically.

Candidate Ranking Flow: After AI interview completion, the platform calculates weighted scores using CV performance, interview results, and integrity penalties. Candidates are ranked automatically for HR review.

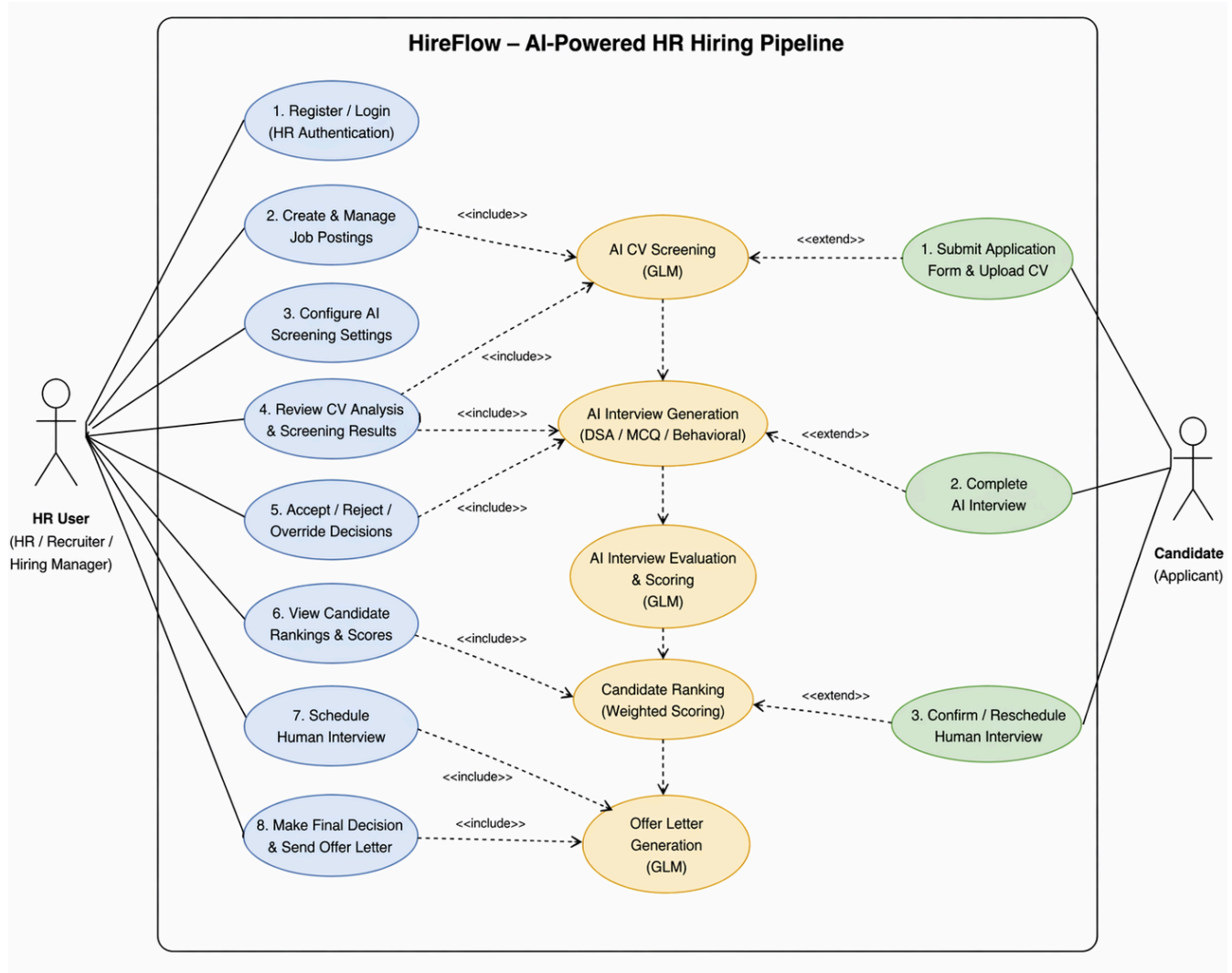
Human Interview Flow: HR schedules the final interview, sends email invitations, and candidates can confirm attendance or request rescheduling through public links.

Final Decision Flow: After the human interview, HR marks the candidate as accepted or rejected. Successful candidates proceed to the offer stage.

Offer Generation Flow: The system generates an AI-assisted offer letter, sends it by email, and updates the candidate status to onboarding or hired status.

Dashboard Monitoring Flow: HR monitors recruitment funnel metrics, job-specific candidate pipelines, ranking tables, and status history through the dashboard.

● **Use Case Diagram:**



6. Features Included (Scope Definition)

The current project scope includes:

- Job creation, viewing, updating, and closing.
- Public candidate application portal.
- CV upload and local file storage.
- Candidate list and candidate detail view.
- GLM-based CV analysis and scoring.
- Auto-screen decision using configurable threshold.
- HR accept, reject, and override actions.
- AI interviews invite workflow with unique tokens.
- Candidate AI interview intro, room, and completion pages.
- AI-generated DSA, MCQ, and behavioral interview questions.
- Candidate answer submission.
- GLM-based answer evaluation.
- Proctor event logging.
- Screen and camera recording upload support.
- Weighted AI interview score and candidate ranking.
- Ranked shortlist view for HR.
- Human interview scheduling.
- Candidates confirm and reschedule response pages.
- Offer generation and email sending.
- Dashboard funnel statistics.
- Status history audit log.
- Workflow state machine for candidate progress.
- WhatsApp or Telegram candidate outreach.
- GitHub profile crawler for candidate coding profile analysis.
- Bias audit dashboard and statistical fairness reporting.

7. Features Not Included (Scope Control)

- Advanced admin role management beyond the existing HR/admin enum.
- Enterprise multi-company or multi-tenant support.
- Cloud file storage such as AWS S3 or Google Cloud Storage.
- Payment, subscription, or billing features.
- Full calendar integration with Google Calendar or Outlook Calendar.
- Job Application LinkedIn Post Auto-Generation.
- Automated Onboarding Workflow (meeting rooms, laptops, ID badges).
- Joinly AI Interviewer Integration.
- Complete candidate account system with candidate login.
- Automated background checks or identity verification.
- Production deployment, scaling, and monitoring setup.

8. Assumptions & Constraints

- **LLM Cost Constraint:** HireFlow uses Z AI GLM for core AI features such as CV screening, interview generation, candidate answer evaluation, ranking assistance, and offer letter drafting. The estimated token cost per average user session remains moderate because AI calls are triggered only during specific workflow stages rather than continuously. For example, screening occurs only when a CV is submitted, while interview scoring occurs only after responses are completed. To keep inference costs manageable at scale, the system minimizes repeated requests by storing AI outputs in the database, reusing previous evaluations, and using structured prompts that focus only on relevant candidate/job information instead of sending unnecessary context.
- **Technical Constraints:** The system depends on the availability of the GLM API for intelligent features. If the API is unavailable, AI-driven screening, scoring, and generation functions cannot operate fully. Email notifications rely on Gmail SMTP credentials and proper configuration. The backend requires PostgreSQL with Prisma migrations for database setup. Frontend and backend services currently run as separate local environments, which may require additional DevOps setup for production

deployment. Uploaded CV files are stored locally, which is suitable for prototype use but not ideal for enterprise-scale storage requirements.

- **Performance Constraints:** AI response speed may vary depending on API latency, server load, or prompt complexity. Large CV files or long candidate responses may require preprocessing, truncation, or chunking before submission to the model. Since multiple candidates may apply to the same role simultaneously, batch processing at scale may introduce delays. The current system is optimized for hackathon prototype demonstrations rather than high-concurrency enterprise workloads.
- **User Input:** The quality of AI recommendations depends heavily on the quality of user-provided inputs. Job descriptions must be clear enough for meaningful CV matching, and candidate CVs must contain structured and relevant information. AI interview scoring also depends on the completeness and relevance of candidate responses. The system is not designed to replace human hiring judgment, therefore HR users are expected to review AI outputs and approve final hiring decisions manually.
- **Security & Operational Constraints:** Proctoring is currently event-based, meaning it logs suspicious actions such as tab switching, camera loss, or copy-paste behavior, but it does not guarantee full fraud prevention. As a prototype system, HireFlow prioritizes functional workflow automation over enterprise-grade hardening such as advanced access control, large-scale monitoring, and high-availability infrastructure.

9. Risks & Questions Throughout Development

- **AI Reliability & Response Stability:** How do we ensure the AI model consistently returns accurate, structured, and relevant outputs for CV screening, interview scoring, and offer generation?
- **Fairness & Evaluation Consistency:** How do we ensure AI scoring remains fair and unbiased across different candidates?
- **Workflow State Stability:** What happens if a candidate becomes stuck between hiring stages due to system errors or invalid transitions?
- **Candidate Integrity During AI Interviews:** How do we verify that interview results reflect the candidate's real ability?
- **Email Communication Reliability:** What happens if candidates do not receive interview links, confirmations, or offer letters?
- **File Storage & Scalability:** How will uploaded CVs, interview evidence, and generated reports be managed as usage grows?
- **Privacy & Data Protection:** How should sensitive candidate data such as CVs, recordings, scores, and hiring decisions be protected?
- **Scope Management:** How do we prevent feature expansion from delaying delivery?